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A Review of the Factors and Outcomes of Institutional Interdisciplinary Neuromodulation Committees: A Multi-Center Experience

Scott Pritzlaff, MD, UC Davis, Carmichael, CA, United States, Michael Leong, MD, Stanford University, Redwood City, CA, United States, Mohan Ravi, MD, Stanford University, Jacksonville, FL, USA, Michael Jung, MD, MBA, UC Davis, Sacramento, CA, United States, Vafi Salmasi, MD, MS, Stanford University, Palo Alto, CA, USA, Juliann Cho, MD, UC Irvine, Ravi Prasad, PhD, UC Davis, Matthew Skoblar, DO, UC Davis, Brick Township, NJ, Naileshni Singh, MD, UC Davis, Sacramento, CA, USA

Introduction: To optimize neuromodulation patient selection and management, an institutional interdisciplinary Neuromodulation Committee was created at each of two academic medical centers (University of California, Davis (UCD) and Stanford University). The committee at each institution is composed of physicians focused on advanced interventional treatments, a pain psychologist, and trainees including residents and pain fellows in an ACGME-accredited training program, a nurse practitioner or physician assistant, and a patient service representative. The committee aims to collaboratively optimize neuromodulation candidates, assess medical and psychological risks, and select the best device given a patient's pain condition. In this study, we present the methodology and outcome data of the Neuromodulation Committee at the two institutions.

Methods: After IRB approval, we included all adult patients who were evaluated by the Neuromodulation Committee between 2017-2020 at two academic pain clinics. Patients with insufficient data were excluded from the study. A retrospective chart review was completed on 385 UCD and Stanford University patient committee reviews. Data collected from the chart review included demographics (age, gender, race, ethnicity, height, weight, BMI), committee meeting results (proceed with trial/implant or decline), and trial success and implant rate.

Results: Out of the 385 screen patients, the committees recommended proceeding with an implantable device (peripheral and neuraxial) in 337 patients (87.5%), whereas continuation of conservative therapy was deemed more appropriate than neuromodulation for the remaining 48 patients (12.5%). In these 48 patients, psychological comorbidities were the main barrier to advanced interventions in 23 patients (47.9%). The remaining patients in this group had a variety of other medical factors making implantable devices a less appropriate treatment path. Of the 278 patients recommended for neuraxial neuro-stimulation, 131 underwent trials with percutaneous leads (47.1%). Trials were successful (resulting in at least a 50% reduction in self-reported pain or improved function) in 108 patients (82.4%). The institutions completed 87 implants out of 108 successful trials.

Conclusion: Even with conscientious selection, one in eight patients reviewed by the committee are deemed that conservative management is recommended over neuromodulation. Patients that were recommended for neuromodulation had high trial to permanent ratios. A proportion of patients did not move forward with a trial of implant procedure even when it was recommended, representing an area for clinical workflow improvement. Ongoing study is needed to explore additional outcomes such as the effectiveness of conservative therapies recommended by the committee, device explant rate, and patient satisfaction.

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Identifying Off-Target Activation Pathways for Baroreflex Activation Therapy (BAT) in Swine

Ashlesha Deshmukh, MS, University of Wisconsin at Madison, Madison, United States, Kevin Cheng, PhD, University of Wisconsin at Madison, Bruce Knudsen, University of Wisconsin at Madison, Megan Settell, BS, PHD, University of Wisconsin at Madison, Maria Laluzerne, University of Wisconsin at Madison, Kathryn Turk, Duke University, Constantinos Tsiptsis, Case Western Reserve University, Ricardo Sanchez, University of Wisconsin at Madison, Kshirin Anand, University of Wisconsin at Madison, Andrew Shoffstall, PHD, Case Western Reserve University, Kip Ludwig, PhD, University of Wisconsin at Madison, Middleton, MN, United States, Nicole Pelot, PhD, Duke University, Durham, NC, United States, Warren Grill, PhD, Duke University, Durham, NC, United States

Introduction: Baroreflex activation therapy (BAT) is FDA approved to treat drug-resistant hypertension with ongoing clinical trials for heart failure [1]. In BAT, electrical stimulation activates baroreceptors in the carotid sinus bulb that function to sense changes in blood pressure (BP). The transduced changes in BP are then transmitted to the brain via the glossopharyngeal nerve, and this signal ultimately decreases sympathetic and increases parasympathetic outflow to reduce BP. However, efficacy and patient responder rate are limited by side effects including numbness in the head and neck, altered speech, respiratory constriction, dry cough, vomiting, altered sensory and motor function of tongue, and altered sensory function of pharynx and oropharynx [2]. Our goal was to identify the muscle sources of these side effects and their nerve activation pathways.

Methods: Domestic swine were used in this work as the diameter and thickness of the swine carotid artery are closer to human than are those of the dog [3]. A 1-mm monopolar disk electrode—similar to the clinical CVRx NEO electrode—was implanted on the dorsal side of the carotid artery at the bifurcation in the notch, caudal to the carotid apex. Dose-response curves for the drop in BP were calculated by testing two combinations of pulse widths (250, 350 us) and frequencies (25, 35 Hz freq.) over a range of stimulation amplitudes. Evoked muscle responses (EMG) were recorded from multiple neck muscles, and the corresponding off-target nerve pathways were identified by sequentially transecting different nerves suspected to be responsible for the off-target muscle activation.

Results: Off-target muscle groups and their corresponding motor nerve pathways were identified as cricoarytenoid by recurrent laryngeal nerve, cricothyroid by superior laryngeal nerve, sternocleidomastoid and trapezius by the accessory nerve and tongue by hypoglossal nerve. The constrictor muscle group was also activated but their pathway could not be verified.

Conclusion: We identified multiple sources of therapy-limiting side effects of BAT in the swine model. These results will help guide the design of improved stimulation electrodes, surgical placement, and parameter programming in the clinic to reduce BAT-associated side effects while increasing on-target activation in patients.

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Investigation of EEG Spectral Features in Long-Term Spinal Cord Stimulation Patients

Tomas Swickley, BS, Florida Atlantic University, Boca Raton, United States, Stacey Tevelev, BS, Florida Atlantic University, Olga Khazen, MS, Florida Atlantic University, Boca Raton, FL, USA, Marisa DiMarzio, PHD, Florida Atlantic University, Boca Raton, FL, USA, Julie Pilitsis, MD, PhD, Florida Atlantic University, Boca Raton, FL, United States, Ilknur Telkes, PhD, MS, Florida Atlantic University, Boca Raton, FL, United States

Introduction: Electroencephalogram (EEG) signatures for patients undergoing spinal cord stimulation (SCS) treatment for chronic pain have largely been studied for patients with relatively new implants.¹ Our understanding of cortical changes over the course of treatment for these patients is limited and requires further investigation. Within this study, we aimed to describe the cortical changes patients may sustain after undergoing SCS for an extended period.

Methods: We recorded 60-channel EEGs in 7 patients with chronic pain 2.04 ± 0.53 years (mean $\pm$ SEM). after successful SCS implantation. We processed the data using the custom MATLAB scripts and EEGLAB toolbox, and utilized the Multiple Artifact Rejection Algorithm (MARA) to evaluate components derived from the independent component analysis. We investigated the stimulation OFF (baseline) and stimulation ON (clinically optimized settings) by computing relative subband power in theta (4-8Hz), alpha (8-12Hz), beta (13-30Hz), and gamma (30-70Hz) bands. Additionally, we investigated frequency changes and correlation of spectral features with the patient-reported outcome measures at the day of EEG recoding.

Results: The cohort consisted of five females and two males, who were diagnosed with chronic neuropathic pain (n=2), failed back surgery syndrome (n=3), or complex pain regional syndrome (n=2). The cohort had a mean age of 58.86 ± 3.97 years. Four patients were implanted with percutaneous leads and three with paddle leads at thoracic level. The average Numeric Rating Scale (NRS) score at the day of EEG session was 2.93 ± 0.68 (scale:1-10) and all patients confirmed that they were receiving pain relief. When the patients were grouped into tonic (n=2), HF10 (n=3), and burst (n=3) based on their clinically optimized SCS settings, the relative power spectra (ON-to-OFF) demonstrated slower alpha rhythms in tonic and faster alpha rhythms in HF10 (Figure 1). Burst stimulation indicated comparable alpha rhythms with stronger beta activity. The statistical comparison between groups indicated that relative theta power and relative alpha power in patients with burst and HF10 were significantly different than the ones with tonic stimulation ($\chi^2(2)=107.08$, $p<0.01$; $\chi^2(2)=59.36$, $p<0.01$, respectively).

Conclusion: The trends noted in the EEG features demonstrate that there is a potential relationship between SCS treatment and these changes in cortical features, however this study is limited by its small sample size. The results of the present study point toward a need for future studies to include a larger sample size that could elucidate a significant overall trend for changes in EEG features in long-term SCS patients which can help us to design better neuromodulation therapies.

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